$\boxed{\text{Taylor expansion in 2 symmetric variables}}$

Consider a function in 2 symmetric variables:

$$f(x, y)$$

Symmetry demands that

$$f(x,y) = f(y,x)$$
$$f_x(x,y) = f_y(y,x)$$
$$f_{xx}(x,y) = f_{yy}(y,x)$$
$$f_{xy}(x,y) = f_{xy}(y,x)$$

f is minimized when

$$x + y = 2u_0$$

so we'll expand around $x = u_0$, $y = u_0$

notation:

$$f_x(u_0, u_0) = f_y(u_0, u_0) = k_1$$

$$f_{xx}(u_0, u_0) = f_{yy}(u_0, u_0) = k_2$$

$$f_{xy}(u_0, u_0) = f_{xy}(u_0, u_0) = k_{11}$$

Expanding:

$$f(x,y) = k_1(x - u_0) + k_1(y - u_0) + \frac{k_2}{2}(x - u_0)^2 + \frac{k_2}{2}(y - u_0)^2 + k_{11}(x - u_0)(y - u_0)$$

Define the "symmetric" and "antisymmetric" variables δu and v:

$$x+y-2u_0 = 2\delta u \quad x-y = 2v$$

$$x = \delta u + u_0 + v \quad y = \delta u + u_0 - v$$

In terms of δu and v:

$$f(\delta u, v) = 2k_1 \delta u + \frac{k_2}{2}\left[(\delta u + v)^2 + (\delta u - v)^2\right] + k_{11}(\delta u + v)(\delta u - v)$$

$$f(\delta u, v) = 2k_1 \delta u + k_2[\delta u^2 + v^2] + k_{11}[\delta u^2 - v^2]$$

$$= 2k_1 \delta u + \delta u^2(k_2 + k_{11}) + v^2(k_2 - k_{11})$$

Now, we insisted that $\delta u = 0$ at the free energy minimum!

Let's check.

$$\left(\frac{\partial f}{\partial \delta u}\right)_{\delta u = 0} = 2k_1 = 0 \text{ if } k_1 = 0$$

$$\left(\frac{\partial f}{\partial v}\right)_{v_0} = 2v_0(k_2 - k_{11}) = 0$$ so either $v_0 = 0$ or $k_2 - k_{11} = 0$

$$f(\delta u, v) = (k_2 + k_{11})\delta u^2 + (k_2 - k_{11})v^2$$

Seems fine, right??

But what if we want an expression in terms of $\omega = (\delta u + u_0)^2 - v^2 = xy$

Use the method of "Lagrange multipliers"

must be the case that

$$[(\delta u + u_0)^2 - v^2] - \omega = 0$$

so for any λ

$$\lambda[(\delta u + u_0)^2 - v^2 - \omega] = 0$$

"Lagrange multiplier"

It's no problem to add 0!

$$f(\delta u, v) = 2k_1 \delta u + k_+ \delta u^2 + k_- v^2$$
$$+ \lambda[(\delta u + u_0)^2 - v^2 - \omega] = 0$$

and now we minimize to find λ:

so,

$$\left.\frac{\partial f}{\partial \delta u}\right|_0 = 2k_1 + \lambda[2(\cancel{\delta u} + u_0)] = 0 \Rightarrow \lambda = -\frac{k_1}{u_0}$$

$$\left.\frac{\partial f}{\partial v}\right| = 2k_- v - 2v\lambda = 0 \Rightarrow \lambda = k_-$$

$$\left.\frac{\partial f}{\partial \lambda}\right|_0 = (\delta u + u_0)^2 - v^2 - \omega) = 0$$ ← just the orig constraint

So now we know $\lambda = k_- = -\frac{k_1}{u_0}$

$$f(\delta u, v) = \delta u^2 [k_+ + \overset{k_-}{\lambda}] + \delta u \overbrace{[2k_1 + 2u_0\lambda]}^{0}$$

$$+ v^2 \underbrace{[k_- - \lambda]}_{0} - \underset{\uparrow \atop k_-}{\lambda} \omega + \underbrace{u_0^2 \lambda}_{-k_1 u_0}$$

$$f(\delta u, \omega) = \delta u^2 \underbrace{[k_+ + k_-]}_{2k_2} - k_- \omega - \underbrace{k_1 u_0}_{\text{constant}}$$

$$\boxed{f(\delta u, \omega) = 2k_2 \, \delta u^2 - (k_2 - k_{11}) \omega}$$

note: this expression still has no k_1 in it! But we couldn't enforce $k_1 = 0$ and enforce $\omega = xy = [(\delta u + u_0)^2 - v^2]$